



Capacity Building for Climate Adaptation in Nigeria: A Community-Based, Data-Driven Approach



Ushahidi



NETHOPE



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List of Abbreviations

YARN: Youth in Agroecology and Restoration Network

NGO: Non - Governmental Organization

LGA: Local Government Areas

NBS: National Bureau of Statistics

Executive Summary

This report presents the findings from a climate adaptation project implemented across Ondo, Osun, and Oyo States in Nigeria between July and December 2025. The initiative aimed to surface community-level insights on the lived impacts of climate change, environmental degradation, and social vulnerability, while identifying locally informed adaptation and mitigation strategies.

To collect these insights, the project adopted a hybrid data collection approach using two tools developed by Ushahidi and co-created with community members across all three states. The first was a mobile-based survey platform deployed by community champions from the Youth in Agroecology and Restoration Network (YARN). The second was Distant Voices, Ushahidi’s AI-powered platform that turns real-world conversations into geographically mapped insights—allowing organizations and decision-makers to see, hear, and act on the needs of the communities they serve. In total, the project collected 9,506 community-generated posts, including text submissions, social media insights, and structured survey responses, exceeding the project target of 5,000 posts.

Key findings highlight significant and interlinked impacts of climate change, including erratic rainfall, shortened planting seasons, rising temperatures, flooding, and soil degradation. These environmental stressors have contributed to crop failure, food insecurity, increased caregiving burdens, disrupted education, and heightened pressure on shared community resources. The health effects, especially heat-related stress and the spread of waterborne diseases, were frequently raised by residents, with limited access to health facilities compounding the challenge.

Community members also proposed a wide range of coping strategies, from staggered planting and early harvesting to communal tree planting and the use of water storage tanks. However, these efforts often remain informal and unsupported by systemic climate adaptation infrastructure or policy. The report further draws on insights from a multi-stakeholder validation meeting, where key adaptation priorities and policy gaps were collaboratively reviewed with local government officials, researchers, and civil society actors.

The project demonstrates that community intelligence, when surfaced systematically and inclusively, can inform climate policy, strengthen early warning systems, and support locally grounded resilience strategies. It calls for greater investment in participatory approaches that bridge community realities with national and subnational climate response frameworks.

Project Partners

The Climate Action Project in Nigeria was implemented through a strategic partnership that combined funding, technology, and local community engagement. [NetHope](#), an international non-profit consortium of humanitarian and development organizations, served as the project funder, providing financial support and strategic oversight. [Ushahidi](#), a global not-for-profit technology organization, delivered the technology platform and led the design and implementation of digital data collection tools, enabling real-time community reporting and analysis. [YARN \(Youth in Agroecology and Restoration Network\)](#), an NGO registered in Nigeria, provided critical on-the-ground support, leveraging its long-standing presence in the project locations and established relationships with local communities to facilitate outreach, engagement, and trust-building. This collaborative partnership ensured that the project combined global technical expertise with locally grounded implementation, enhancing both data quality and community participation.



Introduction

► Location and Administrative Units

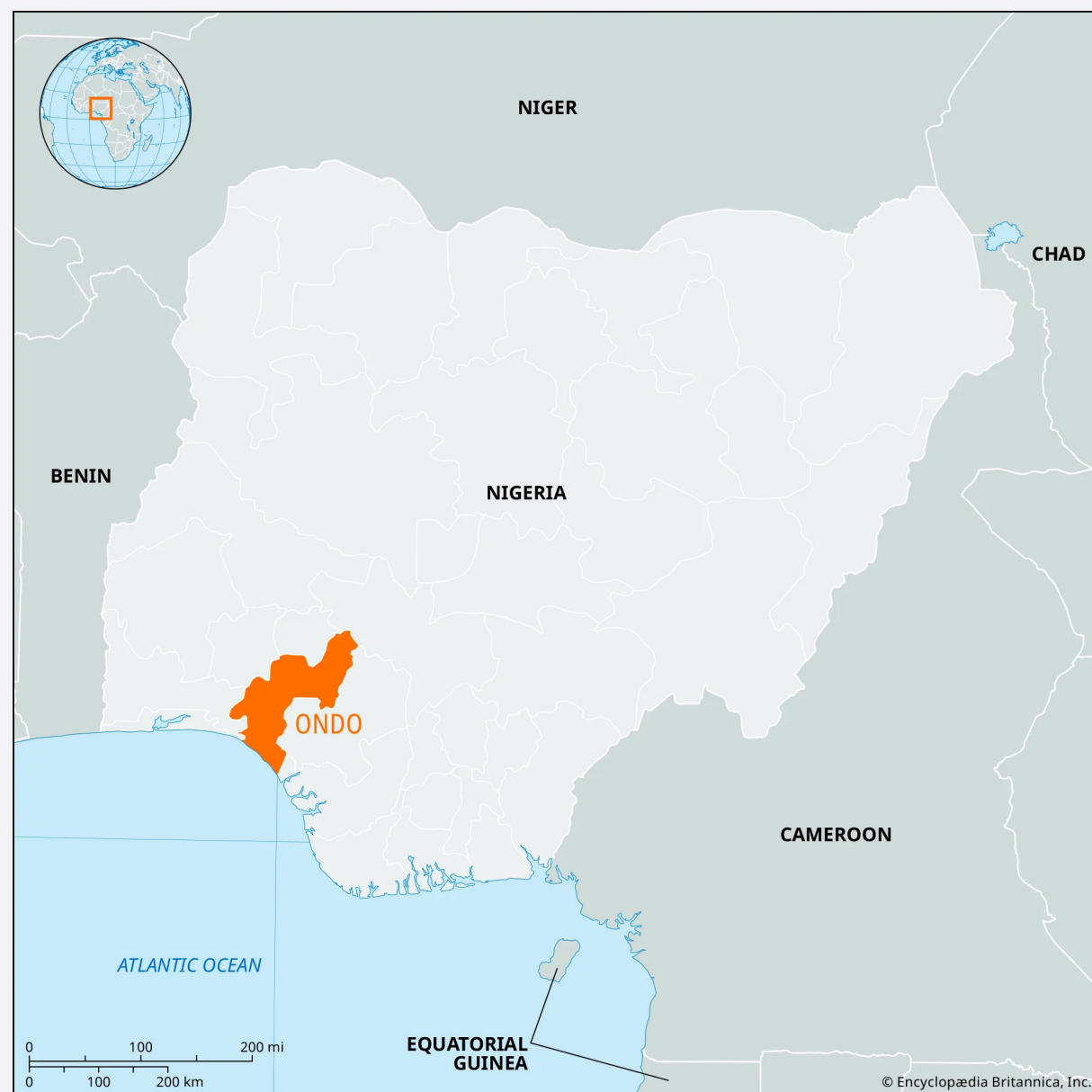
Oyo, Osun, and Ondo States are located in the southwestern region of Nigeria, each forming part of the Yoruba cultural and linguistic area.



Oyo State shares boundaries with Kwara State to the north, Osun State to the east, and Ogun State to the south, while its western boundary touches the Republic of Benin. The state capital is Ibadan, which is one of the largest cities in West Africa and a major centre of commerce, education, and administration. Administratively, Oyo State is divided into 33 Local Government Areas (LGAs), including prominent ones such as Ibadan North, Ibadan South-West, Ogbomosho North, Oyo East, and Ibarapa Central.



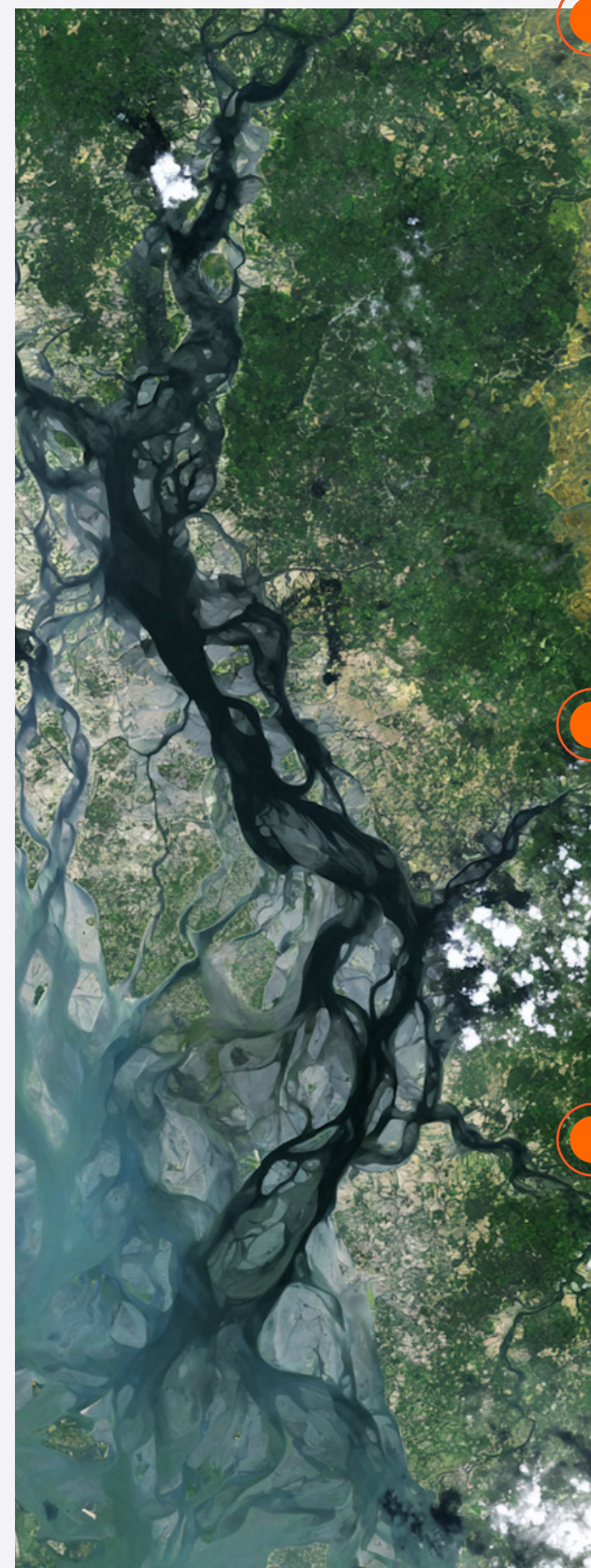
Osun State lies to the south of Oyo State, sharing borders with Kwara State to the north, Ekiti and Ondo States to the east, and Ogun State to the south. Its capital, Osogbo, serves as both the administrative and cultural hub of the state. Osun is famed for its rich Yoruba heritage and annual Osun-Osogbo Festival, a UNESCO-recognized cultural event. The state is made up of 30 Local Government Areas (LGAs), which include Osogbo, Ife Central, Ilesa East, Ede North, and Iwo, among others. These LGAs are organized into zones such as Osogbo, Ife/Ijesha, and Iwo, which facilitate administrative coordination and development planning across the state.



Ondo State lies to the southeast of Osun State. It shares boundaries with Ekiti and Kogi States to the north, Edo State to the east, Delta State to the southeast, Osun State to the west, and the Atlantic Ocean to the south, giving it both inland and coastal characteristics. The state capital is Akure, which serves as the administrative and commercial centre. Ondo State consists of 18 Local Government Areas (LGAs), including Akure South, Ondo West, Owo, Okitipupa, and Ilaje.

Together, Oyo, Osun, and Ondo States play a vital role in Nigeria's socio-economic and cultural landscape. They share deep-rooted Yoruba traditions, a history of precolonial kingdoms, and a strong emphasis on education, commerce, and agriculture. Their administrative units reflect Nigeria's federal system, ensuring decentralized governance and localized development across urban and rural communities.

► Topographic and Drainage Features



Oyo's landscape is marked by broad plains and modest uplands with a network of seasonal rivers; Osun's is defined by rolling hills and a cohesive river system dominated by the Osun River; while Ondo combines coastal swamps with northern highlands, creating a diverse topographic and drainage environment that transitions from inland plateaus to the Atlantic shoreline. Oyo State lies predominantly within the southwestern highlands and is characterized by gently undulating plains interspersed with low hills. The land rises gradually from the south towards the north, with average elevations ranging between 150 and 360 meters above sea level. The terrain is highest in the northwestern and central parts, particularly around Saki, Iseyin, and the Old Oyo area, where granite outcrops and rocky ridges mark the landscape. The southern portion, including Ibadan and Ogbomosho, is lower and more gently sloped. The drainage of Oyo State is dominated by a network of rivers that flow southward and eastward, forming part of the Ogun and Osun river basins. Notable among these are the Ogun, Ona, and Ogbere Rivers, which have numerous tributaries and exhibit seasonal flow—swollen during the rainy months and reduced during the dry season.

Osun State, situated to the east of Oyo, displays a more rolling terrain, with elevations between 200 and 600 meters. It is part of the Yoruba uplands, underlain largely by ancient basement complex rocks. The landscape consists of gentle hills and ridges, particularly around Oke-Ila, Ilesa, and the Ife area, and slopes gradually southwards. The state's drainage pattern is largely dendritic, typical of basement rock regions. The Osun River, from which the state derives its name, is the main watercourse, originating near Ilesa and flowing southwestward until it joins the Ogun River before reaching the Lagos Lagoon. Other rivers such as the Erinle, Oba, and Omi contribute to the hydrological network, supporting agriculture and providing water for domestic use.

Ondo State presents a marked contrast, with topography ranging from coastal lowlands in the south to rugged uplands in the north. The southern part, particularly around Ilaje and Ese-Odo, lies at sea level and is dominated by extensive swamps, creeks, and mangrove forests along the Atlantic coast. Moving northwards, the land rises steadily towards the Akoko and Owo highlands, where elevations reach up to 600 meters. These uplands are part of the Yoruba Hills and are characterized by rocky outcrops and steep slopes. Ondo's drainage system reflects its varied terrain: in the interior, rivers such as the Ogbese, Owena, and Oluwa flow southward in dendritic patterns, while the southern coastal belt features tidal and lagoon systems, including the Mahin Lagoon and numerous estuaries that open into the Atlantic Ocean.

► Climate and Ecological Conditions



In Oyo State, the climate is largely tropical, with two major seasons, the rainy season, which lasts from April to October, and the dry season, from November to March. The state enjoys moderate to high rainfall, with annual averages ranging from about 1,200 mm in the northern parts to over 1,800 mm in the southern areas. Temperatures are generally warm throughout the year, averaging between 25°C and 35°C. Ecologically, Oyo State lies within the forest–savanna transition zone. The southern part, including Ibadan and its environs, is covered with secondary rainforest vegetation, while the northern areas around Ogbomosho and Saki gradually give way to derived savanna, characterized by scattered trees and tall grasses. This diverse ecology supports agriculture, particularly the cultivation of crops such as maize, cassava, cocoa, and yams.

Osun State experiences similar climatic conditions, marked by abundant rainfall and high humidity. The rainy season typically extends from March or April to October, with an annual rainfall of between 1,400 mm and 2,000 mm. The dry season, which follows, is shorter and is accompanied by the Harmattan—a cool, dusty wind from the Sahara that blows between December and February. Temperatures generally range from 21°C to 32°C. Ecologically, Osun State falls within the tropical rainforest zone, endowed with thick vegetation, fertile soils, and numerous rivers and streams, including the Osun River, which holds cultural and spiritual significance. This lush environment favors the cultivation of food and cash crops like cocoa, plantain, oil palm, and kola nut, and it also supports forest resources such as timber and medicinal plants.

In Ondo State, the climate is equally tropical but varies slightly due to its location, extending from the forest belt in the north to the coastal region in the south. The rainy season usually spans from March to November, while the dry season occurs from December to February. The southern coastal areas experience higher humidity and heavier rainfall, averaging up to 2,500 mm annually, while the northern parts receive slightly less. Temperatures generally range from 24°C to 33°C throughout the year. Ecologically, Ondo State encompasses diverse vegetation zones, including dense tropical rainforest, mangrove swamps along the coastal belt, and patches of derived savanna in the northern areas. This variation supports a wide range of economic activities, from fishing and oil exploration in the coastal areas to extensive cocoa and rubber farming in the inland forests.

Overall, the three states share a warm, humid climate that supports rich biodiversity and agricultural productivity. Their ecological zones—ranging from rainforests to derived savannas and coastal wetlands—play a crucial role in shaping their economic activities, settlement patterns, and natural resource distribution.

► Population Size and Composition

Oyo, Osun and Ondo are three neighbouring states in southwestern Nigeria with predominantly Yoruba populations but different population sizes and urban profiles. Official projections from the National Bureau of Statistics (NBS) and publicly available state projections place Oyo as the largest of the three, while Ondo and Osun are smaller. [According to the 2006 National Population Census, Osun State had a population of 3,416,959, which was projected to reach 5,638,559 by 2022 and is expected to rise significantly to 13,977,784 by 2050. In 2006, Ondo State recorded a population of 3,460,877, comprising 1,745,057 males and 1,715,820 females. Oyo State's estimated population currently stands at 8,183,356.](#)

Age & gender



Nigeria's last full national census was in 2006; most state age/sex distributions used by planners are derived from that census and projected forward. At state level, up-to-date published age pyramids are rare; planners therefore rely on the National Bureau of Statistics (NBS) projections and national age-structure patterns (young median age, high share under 15). Expect a youthful population profile in all three states (large cohorts under 25), mirroring the national structure.

Oyo has the largest urban concentration among the three because it contains Ibadan, one of Nigeria's largest cities and West Africa's major urban centres, making Oyo relatively more urbanised and with higher internal urban density.

Osun and Ondo have important urban centres (Osogbo in Osun; Akure and Ondo town in Ondo), but a larger share of their populations live in semi-urban and rural Local Government Areas (LGAs) compared with Oyo.

Population density



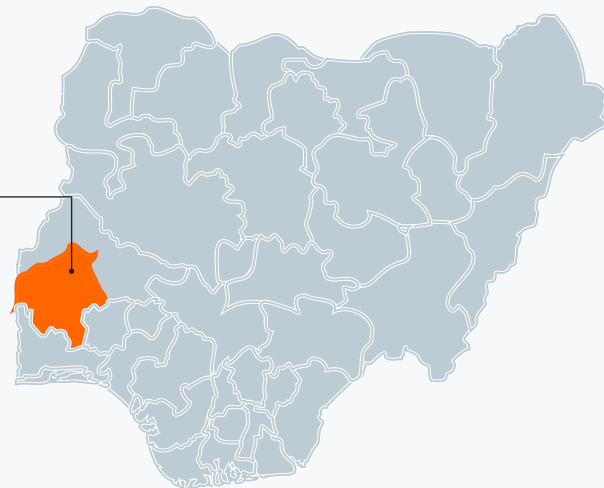
Density varies by LGA and between the three states: Oyo's overall area and large urban population yield high density in Ibadan metro LGAs; Osun is relatively smaller in area with moderate to high density in its central LGAs; Ondo has lowland coastal LGAs (lower density) and higher density around Akure and Ilaje/Okitipupa corridors.

Ethnicity & language



All three states are predominantly Yoruba in ethnicity, and Yoruba language variants are dominant; there are local subgroups and dialectal differences (e.g., Akoko, Akure, Ijesha, Oyo subgroups). Religious composition mixes Christianity, Islam, and traditional beliefs in varying proportions by locality.

Disaster Profile



Oyo State



Climate and Hazard Context

Across [southwestern Nigeria](#), climate change has already resulted in increased temperature, irregular rainfall, and more frequent extreme events like flooding and drought. These trends have been documented in national and regional climate assessments.

Flood and Rainfall Variability: Oyo State regularly experiences intense rainfall events that can lead to urban and riverine flooding, particularly in and around Ibadan and other low-lying areas. Heavy rainfall events are noted for their potential to damage infrastructure, disrupt transport, and impact agriculture. Rainfall patterns have become more unpredictable, with delayed onset and inconsistent distribution across the wet season—effects commonly associated with regional climate change.

Drought and Heat Stress

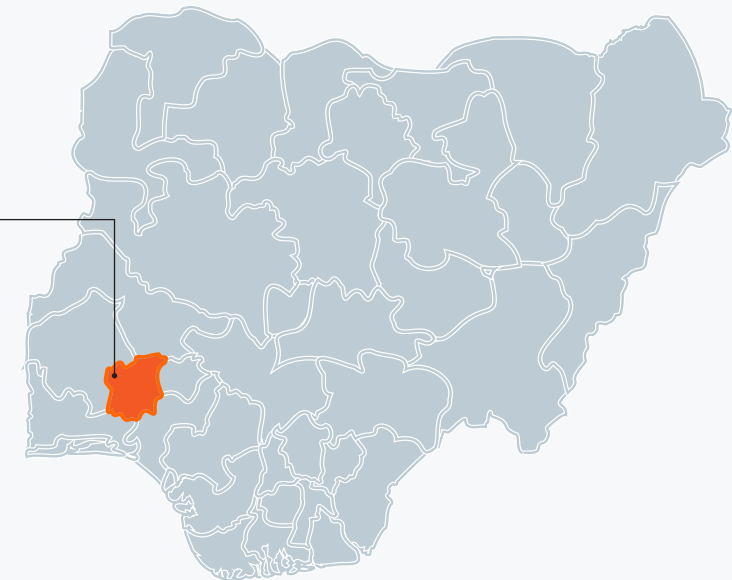


Studies in Oyo show that [smallholder farmers](#) perceive longer dry periods, increased temperatures, and delayed rains, which compromise planting cycles and soil moisture. These patterns reflect broader trends in climate variability affecting agricultural productivity in the state. Heat extremes are increasingly significant across Nigeria, contributing to crop stress and reduced yields — a pattern likely mirrored in Southwest states, including Oyo.

Agricultural and Livelihood Impacts



The combination of erratic rainfall and [rising temperatures](#) adversely affects crop yields, soil fertility, and planting decisions among Oyo's predominantly rain-fed farming populations. Climate change has been shown to affect food security and agricultural livelihoods by reducing water availability and increasing the incidence of crop failure.



Osun State



Climate Context

Osun State shares much of the climatic and agricultural setting of the Southwest, with a largely agricultural economy dependent on rain-fed systems. Across this region, climate change has been linked to variable precipitation, extreme weather events, and shifts in seasonal patterns.



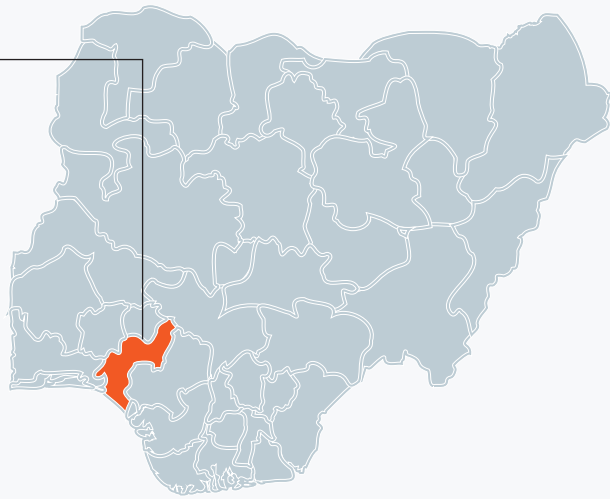
Flood Risk and Variability

Like Oyo, Osun experiences [seasonal flooding](#) during peak rains, which can damage crops, infrastructure, and settlements. The increasing intensity of rainfall events has heightened flood vulnerability in riverine and peri-urban areas. Variability in rainfall timing can shorten effective growing seasons and complicate agricultural planning.



Rural Vulnerability

Rural communities in Osun are highly dependent on agriculture and are therefore sensitive to [climate variability](#). Erratic rainfall and heat stress negatively affect both crop and livestock production, with corresponding socio-economic impacts on rural livelihoods.



Climate and Coastal Context



Ondo State's geography includes inland farmland and coastal/riverside communities, making it vulnerable to both [rainfall extremes](#) and coastal hazards like erosion and sea-level changes. Nigeria's coast is recognized as susceptible to sea-level rise, which contributes to shoreline instability, saline intrusion, and flood risk in low-lying coastal areas.

Rainfall and Flooding



Inland parts of Ondo face flash flooding and runoff hazards during intense rainfall, intensified by urban growth and limited drainage infrastructure. These flooding patterns affect roads, markets, and agricultural land, particularly in and around Akure and southern LGAs.

Coastal and Environmental Hazards



Coastal areas of Ondo, including riverine communities, are exposed to erosion and shoreline change driven by long-term environmental processes and extreme tidal events, a consequence of climate variability and coastal dynamics. Such coastal processes disrupt fishing livelihoods and can degrade settlement infrastructure.

Summary of Hazards by State

State	Primary Climate Hazards	Socio-economic Impact
Oyo	Flooding during the wet season, drought & heat variability	Crop loss, soil stress, heat stress, water scarcity
Osun	Rainfall variability and flooding	Crop disruption, reduced harvests, livelihood stress
Ondo	Inland flooding + coastal erosion	Damage to infrastructure, erosion threats, and disrupted fisheries

Project Objectives and Methodology

► The primary objectives of the project were to:

1. Co-create a Data Collection Framework for Climate-Specific Data

Partner with NetHope and a YARN (Youth in Agroecology and Restoration Network) to co-create a data collection framework that utilizes Ushahidi's Distant Voices technology. The process would begin by curating a long-form questionnaire to serve as a foundation to generate a list of relevant keywords that would be used for social media listening. The keywords would be carefully selected based on the themes and topics identified in the questionnaire, ensuring that they accurately reflect the community's climate context. The list of keywords would then be incorporated into Ushahidi's Distant Voices technology to guide the AI-powered social media listening process and data collection, ensuring that the insights gathered are highly relevant and actionable.

2. Social Media Listening and Data Analysis to Assess Climate Risks and Adaptation Practices

Use Ushahidi's Distant Voices technology to conduct social media listening on Facebook and scrape publicly available data from at least 5,000 unique conversations in Oyo, Ondo and Osun. The listening would cover conversations in English and the Yoruba language. The aim was to assess key climate risks, adaptation needs, and current adaptation practices as identified by the community. The collected data would be analyzed using sentiment analysis and AI-driven categorization to identify trends, successful adaptation practices, and areas where adaptive measures may be lacking. This would provide actionable insights for informing climate resilience strategies, highlighting the most pressing climate challenges and opportunities for scaling up local adaptation efforts.

3. Validate Data and Adaptation Strategies with Community Members

Conduct in-person validation sessions with a diverse cross-section of community members (youth, women, elders, community leaders, persons with disabilities, etc.) in Oyo, Ondo, and Osun. The aim was to verify the climate risks and adaptation strategies identified through the data analysis. The sessions would provide an opportunity to gather community feedback on the accuracy, relevance, and applicability of the adaptation strategies derived from the data. The community's input would ensure that the proposed solutions are practical, culturally appropriate, and aligned with local needs. The process would also empower the community by giving them ownership over the adaptation strategies that directly impact their livelihoods.

4. Disseminate Adaptation Findings and Foster Collaborative Action

Share the findings of the data collection and community insights through Ushahidi and NetHope platforms, ensuring that the results are accessible to a wide range of stakeholders. In addition, the findings would be directly shared with local government authorities in Oyo, Ondo and Osun States to ensure that the data informs local decision-making and climate adaptation policies. The goal is to increase awareness of climate risks and adaptation strategies, fostering collective action among community leaders, local government officials, and civil society to implement climate-resilient practices across the three states. This would ensure the findings not only reach the community but also help shape policy and guide governance at the local level.



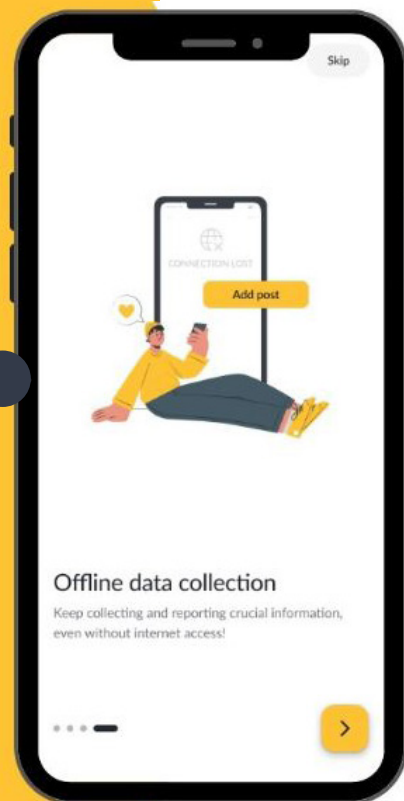
- The project employed a dual data-collection approach to ensure broad, inclusive, and geographically representative participation across Oyo, Osun, and Ondo States. Data was collected through:

1. Mobile App

Youth in Agroecology and Restoration Network (YARN) led in-person and community-based data collection using the Ushahidi mobile application. Trained data champions worked directly within communities to collect responses, ensuring participation from individuals who may have limited access to digital platforms or internet connectivity. They engaged directly with community members, enabling real-time submissions of climate-related experiences, observations, and risks.

This approach supported:

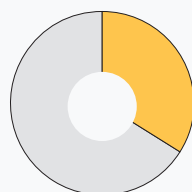
- Structured survey administration
- Inclusion of semi-urban and rural populations
- Gender- and age-diverse participation
- Improved data quality through guided facilitation



Submissions collected through the Ushahidi Platform (mobile app) using this approach accounted for 5,954 posts, representing 63% of total engagement. Each state, Oyo, Osun, and Ondo, was almost equally represented on the platform:

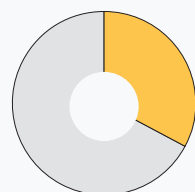
Ondo State:

34%



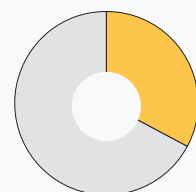
Oyo State:

33%



Osun State:

33%



2. Ushahidi - Distant Voices



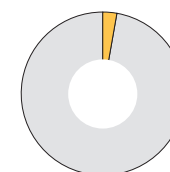
Ushahidi complemented the direct data collection with Distant Voices, a tool that mines climate-related discussions from social media platforms. This included [Nairaland](#), X, and Facebook. A total of 3,552 posts were curated from these sources, representing 37% of total engagement. This approach enabled the project to tap into community sentiment, trending concerns, and local discourse that were beyond the reach of physical data champions. Specifically:

- Capturing spontaneous, citizen-generated insights
- Enabling participation beyond organized data collection sessions
- Gathering perceptions, narratives, and community discourse around climate impacts

These submissions were sourced from:

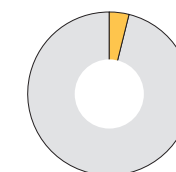
Facebook:

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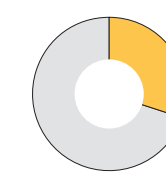
Twitter:

4%



Nairaland:

30%



Value of the Hybrid Engagement Model

The combination of field-based mobile data collection (YARN) and remote reporting (Ushahidi Distant Voices) strengthened the overall evidence base by:

- Reaching communities with varying levels of digital access (both connected and offline)
- Capturing both structured survey responses and unprompted community narratives
- Enabling triangulation between lived experiences, perceptions, and reported needs
- Enhancing the credibility and representativeness of findings used for analysis and recommendations

This hybrid approach ensured that the project did not rely on a single data stream, but instead reflected multiple modes of participation, reinforcing the legitimacy of the insights presented in subsequent sections of the report. Together, these approaches generated a total of 9,506 community submissions, reflecting both structured, field-based engagement and remote, citizen-led reporting.

► Contextual Insights from Pre-Implementation Discovery Sessions

Held from 30th June to 2nd July, the discovery sessions conducted across Ondo, Osun, and Oyo States served as foundational engagements to surface local knowledge, validate the relevance of the proposed data collection tools, and ensure alignment with community priorities. These sessions brought together government stakeholders, civil society organisations, media, youth, and representatives from rural and semi-urban communities, providing deep insights into climate perceptions, lived realities, and ongoing adaptation efforts.

► Purpose of the Discovery Sessions

The discovery sessions were a collaborative effort led by Ushahidi, NetHope, and YARN (Youth in Agroecology and Restoration Network). The sessions aimed to:

- Introduce the project to community and state-level stakeholders.
- Create an open platform for sharing localised experiences of climate impacts.
- Train the data champions on the use of Ushahidi's mobile App.
- Validate and refine the participatory data collection tools.
- Establish community buy-in and identify context-specific adaptation priorities.
- Encourage collaboration between civil society, government, and community actors in climate governance.

These dialogues were essential in grounding the project in the realities of the communities it intended to serve.

Photos from the discovery sessions:



Key Findings



Cross-Cutting Themes Across the Three States

While each state presented unique contexts and challenges, several common themes emerged from all discovery sessions:

Widespread Climate Disruptions

Participants across all states reported worsening climate conditions, including:

- Increased temperatures and heat waves.
- Erratic rainfall patterns leading to flooding and dry spells.
- Sea incursion and coastal erosion in riverine areas, particularly in Ondo.
- Extended harmattan periods, especially affecting Oyo communities.

These disruptions were cited as significantly affecting farming, fishing, housing, and daily livelihoods.

Low Levels of Awareness and Institutional Support

Stakeholders across Osun, Ondo, and Oyo expressed concerns about:

- Limited public awareness of climate change causes and mitigation.
- Absence or weakness of climate response institutions at local levels.
- Inadequate extension services, particularly for farmers and fisherfolk.
- Poor communication channels for weather warnings and adaptation messaging.

Disproportionate Impact on Vulnerable Groups

All three states flagged the heightened vulnerability of:

- Women, due to caregiving roles and resource dependency.
- Children and the elderly, due to limited mobility and health challenges.
- Rural dwellers whose livelihoods are dependent on natural ecosystems.

Local Adaptation Efforts Already Underway

Despite institutional gaps, communities have initiated their own responses:

- Tree planting campaigns led by youth groups and religious leaders.
- Drainage clearing to reduce urban flooding.
- Climate education through radio and informal community forums.
- Mulching and shifting planting seasons to adapt to rainfall unpredictability.

▶ Insights from State-Level Sessions

Ondo State

Participants highlighted climate issues specific to coastal communities, including:

- Flooding, sea incursion, and forced displacement of communities such as Ayetoro.
- Oil exploration and weak enforcement of environmental laws as aggravating climate vulnerabilities.
- Community members voiced strong support for the project and reviewed the draft questionnaires with enthusiasm.
- State officials acknowledged efforts such as toll-free reporting lines, flood awareness campaigns, and school-based Green Clubs.

Oyo State

The Oyo session, held in Ibadan, brought together representatives from the Ministry of Environment, agriculture officers, youth leaders, and community-based organisations:

- Participants cited erratic rain patterns, extreme heat, and late onset of rains as key concerns.
- Youth participants discussed informal innovations such as composting and climate talk shows on radio.
- Feedback on the draft survey tool emphasized the need for translation into local dialects and simplification for respondents with lower literacy.

Osun State

In Osogbo, stakeholders raised concerns around:

- The failure of previous government interventions in irrigation infrastructure.
- Low-level farmer education on soil preservation and erosion control.
- Some participants expressed initial skepticism about “yet another project” but shifted their views after deeper discussion on the co-creation approach and long-term policy linkages.
- Specific mention was made of youth-led green initiatives and the need for state-level climate adaptation councils.

▶ Value of the Discovery Sessions

The sessions provided:

- Crucial validation of the project's participatory methodology.
- Localised framing of climate issues that shaped the language and priorities of the final data tools.
- Early political and community buy-in, which contributed to the successful rollout of the data collection and storytelling phases.

The discovery sessions were not just pre-project formalities, but catalytic events that grounded the initiative in lived realities, strengthened legitimacy, and laid the groundwork for responsive, community-driven climate governance.

▶ Climate Change Impacts: Shifting Seasons and Agricultural Instability

Across the three states, community members consistently reported the increasing unpredictability of weather patterns. Farmers described a disconnect between their generational knowledge of planting and harvesting seasons and the actual onset of rainfall. What was once a dependable seasonal cycle has become erratic, with rains arriving too late, stopping too early, or being concentrated in damaging bursts. One participant from Oyo lamented: “We plant when we used to, but the rains now come and go as they please. Our crops suffer.”

The consequences of these changes are visible in reduced yields, crop failures, and food insecurity. Staple crops such as cassava, yam, and maize have suffered the most, with many respondents indicating that their traditional planting schedules are no longer effective. In Osun, a participant noted: “I had to plant maize twice this year. The first time it dried up before sprouting.”

In Ondo, some farmers have begun experimenting with staggered planting or shorter-cycle crops, but these adjustments are often made without reliable climate information or extension services. Others, particularly women and elderly farmers, expressed fatigue and frustration at their inability to adapt: “I am too old to keep guessing when to plant. We need help.”

These accounts reflect not only the tangible impacts of climate change but also a deepening sense of uncertainty and helplessness, especially among smallholder farmers who depend solely on rain-fed agriculture.

▶ Climate Change and Health: Heat Stress, Waterborne Diseases, and Compounding Risks

The intersection of climate impacts and public health emerged strongly in participant responses, though it remains largely under-acknowledged in formal planning processes. Rising temperatures, prolonged heat exposure, poor air quality, and stagnant water following floods were consistently cited as drivers of new or worsening health conditions.

Several respondents reported heat-related illnesses, including fatigue, headaches, skin irritation, and heat rashes, particularly among individuals working outdoors. Others noted an increase in skin diseases, which they associated with prolonged exposure to heat, poor water quality, and unhygienic conditions following flooding. In Ondo, one respondent noted: “We can no longer stay long in the sun. The heat is too much.” Another added: “The smoke from garbage burning gives me headaches and makes it hard to breathe.”

The impacts on children and older persons were frequently highlighted. Respondents in Osun linked stagnant floodwaters to malaria outbreaks, especially during the rainy season: “After the flood, we fall sick more. Mosquitoes increase.” Some community members also attributed early menopause among women to prolonged heat exposure and climate-related stress, a perception that warrants further medical and scientific investigation.

Access to healthcare emerged as a compounding challenge. In flood-affected areas, damaged roads and limited transportation made it difficult to reach health facilities even as symptoms worsened. In Oyo, one respondent noted: “The clinic is far. When the rain floods the bridge, we wait and hope we recover.”

Despite these clearly articulated risks, there is limited evidence that local health systems are systematically integrating climate-related risk assessments or community-led health monitoring into planning and response mechanisms. The findings point to an urgent need for localized climate–health action, including early detection of climate-sensitive illnesses, targeted public health outreach, strengthened surveillance, and improved health-system preparedness.

► Social and Community Impacts

Climate change has not only altered environmental conditions and livelihoods in Ondo, Oyo, and Osun States, but it has also deeply affected the social fabric and daily rhythms of community life. Across rural and urban settlements alike, respondents consistently highlighted the compounding social consequences of environmental stressors from disrupted education to increased caregiving burdens and strained family dynamics.

The interruption of children’s schooling emerged as a common theme. When rains intensify beyond seasonal norms or persist for days, floodwaters transform dirt roads into impassable barriers. One parent shared how flooding directly interfered with education: “Flooded roads stop our children from going to school. Sometimes, the school itself is flooded.” This reflects not only lost instructional time, but also a broader sense of community helplessness when basic service access is compromised.

Beyond schooling, the burden of care has shifted disproportionately onto women and girls. Many respondents described spending longer hours collecting water due to drying or contaminated sources, increasing the domestic workload, and reducing time for income-generating activities or formal employment. Some shared sentiments like: “We walk further each week just to find water that doesn’t make us sick.” This echoes broader rural African narratives where climate stress amplifies gender inequalities and redistributes household labor in ways that exacerbate economic insecurity.

Community cohesion is also being tested. Where water points dry up or farmlands become less productive, tensions surface, not always violent, but palpable in shared frustrations and competing needs. Although overt conflict was less frequently described in these submissions than in national headlines, the latent pressure on shared resources: land, water, and community infrastructure, was evident in multiple reports and supports wider research showing that climate variability often increases social strain in agrarian societies.

► Environmental Degradation and Resource Pressure

The environment itself, the physical backdrop on which daily life and livelihoods depend, is undergoing profound change. Multiple intersecting forces, both climatic and human-induced, are intensifying degradation in the project states.

One of the most persistent concerns raised by community members was soil degradation and land exhaustion. Farmers in both rural and peri-urban areas described how seemingly familiar plots now produced less than in past seasons, even under similar cultivation practices. “The soil is not what it used to be,” reported one long-term

resident in a mixed agricultural community. This observation aligns with broader research indicating that erratic rainfall patterns and repeated extreme weather events reduce soil moisture, accelerate nutrient loss, and render previously fertile lands less productive.

Another dimension of environmental stress highlighted by residents was the accelerated loss of tree cover and vegetation, a trend driven by both climate change and adaptive choices under economic pressure. In some communities, respondents linked increasing heat and erratic rainfall to diminishing vegetation: “We cut trees to clear land for farming, but now the sun feels harsher.” While this sentiment reflects a coping strategy, it also highlights a dangerous feedback loop: tree loss reduces shade and soil protection, which in turn exposes land to erosion and heat stress. This scenario mirrors national assessments showing that extensive deforestation and environmental degradation are compounding climate risk in Nigeria.

Resource pressure extends beyond soil to include water scarcity and variability. Multiple submissions referenced drying streams and wells that previously provided reliable access. The declines in water availability are not purely seasonal but tied to shifting precipitation patterns reported by residents and confirmed by climate research. Erratic storms and prolonged dry spells both reduce the predictability and reliability of water sources, forcing households into longer trips for basic needs, strengthening the social disruptions described above.

This combined environmental pressure of weakened soils, depleted vegetation, and inconsistent water supply, is not just a backdrop for community struggles; it is reshaping them. Multiple respondents reflected a sense of loss and frustration not only with tangible impacts, but with the perception that the natural environment they relied upon for generations is now unreliable: “The land doesn’t work like it used to.” While those words are simple, they signify a shared recognition of deep ecological change.

► Economic Impacts: Livelihood Stress and Cost-of-Living Pressures

Climate change and environmental degradation are not just physical phenomena, they are rapidly becoming economic crises for affected communities in Ondo, Osun, and Oyo. Repeated crop failures, reduced soil fertility, and erratic rainfall patterns are directly undermining local livelihoods, particularly for smallholder farmers who rely on agriculture for income and subsistence. As one farmer put it: “We farm the same land but harvest less. Inputs cost more, and we sell for less.”

The economic strain extends beyond the farm. In communities where water sources have dried up or become contaminated, households are now paying for previously free resources, either by purchasing water or spending money on transportation to reach distant sources. In Oyo, one respondent lamented: “We now spend money we don’t have just to find clean water.”

Market access and price volatility also emerged as recurrent issues. Flooding and poor road infrastructure have disrupted local supply chains, making it harder for farmers to transport goods and leading to spikes in food prices. Additionally, climate-linked health impacts such as malaria outbreaks and heat-related illness often translate into out-of-pocket healthcare costs, especially where access to public services is limited. As captured in one submission: “When the child is sick, we borrow to take them to the hospital. But the rains caused the sickness.”

Taken together, these dynamics point to a downward spiral of climate-induced economic precarity, where diminished income, higher costs, and environmental shocks create a persistent cycle of vulnerability. Without targeted support such as crop insurance, input subsidies, or reliable market linkages, many community members expressed concern that their livelihoods may not be sustainable for much longer.

► Intergenerational Dynamics: Shifting Roles, Aspirations, and Tensions

The consequences of climate change are unfolding unevenly across generations. Feedback from participants reveals that youth, elders, and middle-aged caregivers experience climate stress differently, with each group facing unique challenges and carrying distinct burdens.

Elderly farmers, for instance, expressed frustration and resignation over their inability to adapt to new weather patterns. Many described how their lifelong knowledge, once reliable for determining planting seasons and agricultural cycles, has become obsolete. “We don’t understand the weather anymore. What we knew no longer works,” said one elderly participant from Ondo. Several older respondents also cited physical limitations that make it difficult to experiment with new practices, access training, or shift to alternative livelihoods.

In contrast, younger people are increasingly distancing themselves from agriculture. While some youth still engage in farming, many see it as a last resort. This shift in aspirations was voiced repeatedly in the Distant Voices data: “Young people don’t want to farm because they see it failing,” said one respondent. Another added: “We want other work, but there are no jobs here.” This signals a broader socio-economic challenge: as the viability of traditional livelihoods declines, youth are left with limited alternatives, fueling migration, underemployment, and social dissatisfaction.

Meanwhile, women and adolescent girls, often the primary caregivers in households, described heightened burdens as a result of climate disruptions. From walking longer distances to fetch water, to caring for sick family members during heatwaves or flood-related illness outbreaks, their responsibilities have intensified. “When the child is sick and water is far, it is the mother who suffers,” one woman explained during the validation meeting.

These generational perspectives reveal both adaptive divergence and relational strain. The disconnect between older and younger farmers, the gendered burden on women, and the changing aspirations of youth all suggest that effective climate action must be tailored not just by geography or livelihood, but also by age, role, and social position. A failure to recognize these dynamics could deepen inequality and leave key groups disengaged from adaptation efforts.

Validation and Dissemination of Local-Level Climate Adaptability and Resilience Strategies

To ensure the credibility and community ownership of the data collected, a series of validation meetings were held in Ondo, Osun, and Oyo states between November 17th–19th, 2025. These meetings brought together a wide range of stakeholders, including ministries of agriculture, environment, and women's affairs, as well as civil society organizations, traditional leaders, farmers' associations, media representatives, and community champions.

The sessions served two key purposes: to validate the data collected through both enumerator-led mobile app surveys and the Distant Voices platform, and to provide a space for stakeholders to interrogate, refine, and co-own the climate insights. These dialogues underscored a high degree of alignment between the captured data and lived experiences on the ground, while also surfacing critical gaps and state-specific nuances. Stakeholders emphasized that the validated data should inform local development plans, disaster risk management efforts, and climate adaptation budgets.

Specific feedback pointed to the urgent need for climate-smart infrastructure, particularly in flood-prone and coastal areas like Ayetoro, and a call for better coordination across government and civil society actors. Ministries across the three states acknowledged the usefulness of the data and committed to integrating it into their programming, from adjusting school hours during heatwaves to revising agricultural extension curricula to include community-proposed adaptation strategies.

Photos from the validation sessions:



Community Feedback and Recommendations

Community Feedback on Environmental Challenges

Temperature and Rainfall Changes

Communities across the three states reported noticeable disruptions to seasonal cycles, citing delayed rainfall onset, unpredictable withdrawal patterns, prolonged dry spells, and extreme heatwaves. These changes have disrupted farming calendars, with one respondent noting: “We used to know when the rains would start and end. Now, we cannot plan. You can plant, and the rain will disappear.” — Farmer, Osun.

Heatwaves were also mentioned as being more intense and frequent, impacting human health, livestock, and even leading to school closures or adjusted hours in some regions.

Distant Voices data mirrored these concerns, especially from urban youth on social platforms: “Omo this heat no be here. People dey faint anyhow. No light, no water, and sun dey fire like say e get beef with us.” — (This heat is unbearable. People are fainting everywhere. There’s no electricity, no water, and the sun is blazing as if it has a personal grudge against us) Twitter user, March 2025

Other Environmental Challenges

Communities highlighted environmental degradation driven by deforestation, mining, and the encroachment of water bodies into settlements. Plastic waste and blocked drainage systems were cited as exacerbating flood risks. Participants from Ayetoro, in Ondo State, vividly described their homes being swallowed by sea surges: “Sea has taken our land. We don’t sleep in peace anymore. Every rainfall is fear.” — Resident, Ayetoro.

In Osun, mining-related water pollution was highlighted, especially in cocoa-growing regions: “Our streams are brown. We can’t drink them again. Even the crops are drying.” — Farmer, Osun

Effects of Environmental Challenges on Sources of Income

The climate disruptions have directly affected livelihoods, particularly in agriculture, which remains the backbone of income for most respondents. Cocoa, maize, and cassava farmers all reported declining yields, while fisherfolk in coastal areas have been forced to migrate or diversify due to changing water patterns.

From the Ushahidi platform data: “No more fish. We wait hours. The water is hot, and fish are dying.” — Fisherman, Ondo

Others shared frustrations over increased post-harvest losses due to unpredictable weather: “You can’t dry cassava anymore. Rain just comes anyhow.” — Woman farmer, Oyo

Community Adaptation Measures

In response to the mounting climate and environmental challenges, communities across Ondo, Osun, and Oyo States are adopting a variety of coping strategies, many of which are self-taught, locally driven, and often resource-constrained.

One of the most common forms of adaptation is the adoption of alternative energy sources, particularly solar power. Social media narratives collected via Distant Voices frequently spotlighted the increasing promotion and use of solar systems. Tweets such as “We install solar energy systems at affordable prices” and “Going solar

in Nigeria is no longer about escaping bills, it’s about surviving heatwaves and blackouts” reflect how solar is viewed not just as a modern amenity, but as a necessary adaptation tool. These shifts are particularly important in areas with unreliable power supply, where heatwaves and fuel shortages compound daily challenges.

On the agricultural front, smallholder farmers are increasingly turning to staggered planting cycles and climate-resilient crop varieties. Although these decisions are often made without formal extension support, they reflect a grassroots understanding of the new climatic reality. Farmers described adapting by watching the skies, learning from neighbours, and shifting planting calendars. One participant noted, “We no longer plant by date, we plant by signs — when the wind changes, or when the first insects arrive.”

Water conservation practices are also emerging in response to prolonged dry spells and erratic rainfall. Community members have reported digging small water reservoirs or storing rainwater, while women and girls, who bear the brunt of water collection duties, are leading informal discussions on how to better share limited water resources.

However, not all communities have the same capacity to adapt. The validation meeting and field data reveal that elderly farmers, low-income households, and women-headed homes face greater barriers to implementing these adaptation measures, often due to a lack of capital or access to reliable information. As one woman in the validation session put it: “We do what we can, but without support, our efforts won’t go far.”

Recommendations for Climate Action

Response Measures (Emergency Preparedness and Disaster Response)

The recurring episodes of flooding, windstorms, and crop failures underscore the need for localized emergency preparedness systems. Communities recommended that local governments set up early warning mechanisms that can reach even remote or semi-literate populations, ideally using radio, voice alerts, and existing religious or community networks. One participant urged: “Tell us before the floods come. We are tired of being surprised.”

There is also an urgent need for first response training and basic equipment, particularly in flood-prone areas. In many cases, households affected by floods or bushfires had no safety nets, no evacuation plans, and no way to access emergency aid. A recommendation echoed across board was the establishment of local disaster response committees, trained, resourced, and connected to local authorities.

Additionally, community members advocated for the development of accessible shelter points during extreme weather, such as dry community halls or school buildings, to protect children, elderly people, and persons with disabilities.

Adaptation Strategies (Building Resilience to Climate Change)

Communities identified information access as a cornerstone of adaptation. There is a growing demand for locally tailored weather forecasts, especially in Yoruba or other native languages. Farmers requested seasonal calendars, printed or broadcast, with visual cues on when to expect rains or heatwaves. “We need to understand the signs,” one respondent said. “Not just on the radio, but explained to us.”

Another vital strategy is the mainstreaming of climate resilience into agriculture, through government-backed input support, training on water-efficient techniques, and support for crop diversification. Distant Voices responses also revealed interest in home gardening and permaculture, which many urban respondents cited as a way to reduce dependence on expensive markets.

Communities also asked for investment in sustainable livelihoods, including vocational training for youths and

women, especially in areas like clean energy, agri-processing, and eco-tourism. These would not only reduce environmental strain but also offer alternatives to charcoal burning and forest clearing.

One of the state government representatives commended the project and indicated that the data collected would be used to identify high-risk areas for climate-induced hazards. They further explained that the insights would inform updates to the state's climate adaptation plans and guide the prioritization of interventions where they were most needed.

Analysis of the data revealed that the majority of the respondents who reported a lack of knowledge or understanding of climate change had attained education up to secondary level. This finding points to a gap in formal climate education at the secondary and lower education levels. As a result, there is a need for a deliberate and structured inclusion of climate change studies within secondary school curricula and below, to strengthen awareness, build foundational climate literacy, and support long-term climate resilience.

Mitigation Measures (Reducing Long-Term Climate Vulnerabilities)

Community voices also acknowledged the role of human activities in worsening climate impacts, particularly deforestation and waste mismanagement. As such, several participants called for community-led reforestation campaigns, especially around schools, churches, and water sources.

Many respondents also highlighted the growing need for waste management education and infrastructure, pointing to the health risks posed by plastic-choked drainage systems. One woman noted, "The rain brings back what we throw away. We are the cause too."

Finally, there was a clear demand for stronger climate governance and youth inclusion in decision-making. Respondents wanted to see local councils holding public dialogues, recognizing youth climate clubs, and involving communities in designing policies: "We are tired of experts who never return. We are here. Let us be part of the solution."

► **Conclusion**

The Nigeria Climate Adaptation Project has illuminated the everyday realities of climate change across Oyo, Ondo, and Osun States, surfacing stories not only of loss and disruption but also of resilience and action. Through a participatory approach that combined Ushahidi's mobile survey tool and the AI-powered Distant Voices technology, the project collected over 9,500 community submissions, offering a uniquely grounded perspective on environmental and socio-economic shifts.

What emerges is a portrait of communities already grappling with the impacts of a changing climate, shifting seasons, crop failures, flooding, water scarcity, heat stress, and responding in real time with locally generated solutions. These include staggered planting, communal drainage efforts, informal alert systems, and alternative livelihoods. However, these responses remain fragmented, under-resourced, and disconnected from formal adaptation systems. Without targeted investment, policy integration, and systemic support, such grassroots resilience risks being overwhelmed by accelerating climate shocks.

The findings surface five key imperatives that should guide future climate action:

First, it is essential to acknowledge local expertise. Community members' observations of drying rivers, erratic rainfall, hardened soils, and increasing heat closely mirror scientific assessments. These lived experiences are not anecdotal, they are a form of vital, place-based intelligence that must inform climate planning, early warning systems, and risk assessments.

Second, the project reveals the need to reinforce intergenerational inclusion. Elder farmers expressed disorientation as traditional planting calendars failed, while younger residents struggled to find viable pathways in agriculture. Bridging this gap means co-creating new adaptation models that draw on the wisdom of experience while empowering innovation and youth-led resilience strategies.

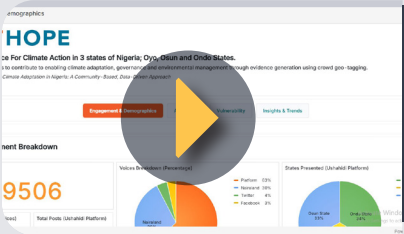
Third, climate change must be treated as a multi-sectoral crisis, not just an environmental issue. The collected stories show clear impacts on health (malaria, heat stress, access to care), education (flooded schools and roads), and the economy (shrinking yields, rising costs, displaced labor). Cross-sector integration of climate responses is therefore urgent.

Fourth, investments in local systems for preparedness and early warning must become standard practice. The data reflects a strong culture of community-driven response, from flood alerts on WhatsApp to farm-level experimentation. These efforts should be formalized and supported through capacity building, linkages to local governments, and inclusion in Nigeria's broader disaster risk reduction frameworks.

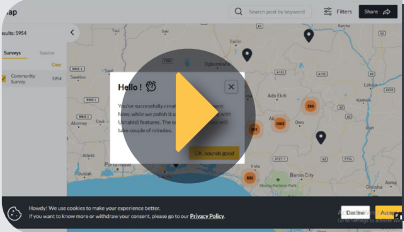
Finally, the success of this project underscores the value of sustaining participatory data systems. Tools like Ushahidi's mobile surveys and Distant Voices enable communities to continuously share, map, and act on their realities. These platforms offer scalable models for real-time feedback loops, accountability, and inclusive governance.

As Nigeria confronts the intertwined crises of climate vulnerability and socio-economic inequality, the voices captured in this report offer a human-centered roadmap. The message is clear: communities are not waiting, they are adapting, organizing, and speaking. It is now the responsibility of government actors, civil society, and development partners to act with urgency, equity, and commitment, ensuring that those most affected by climate change are also at the center of its solutions.

Annexes



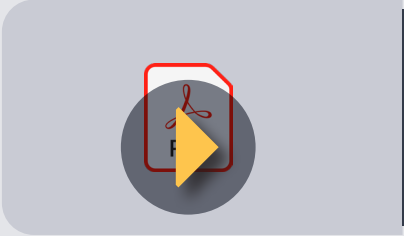
[Data analysis and insights dashboard](#)



[Ushahidi deployment](#)



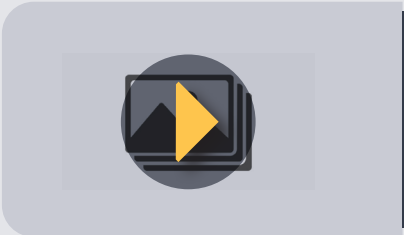
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